



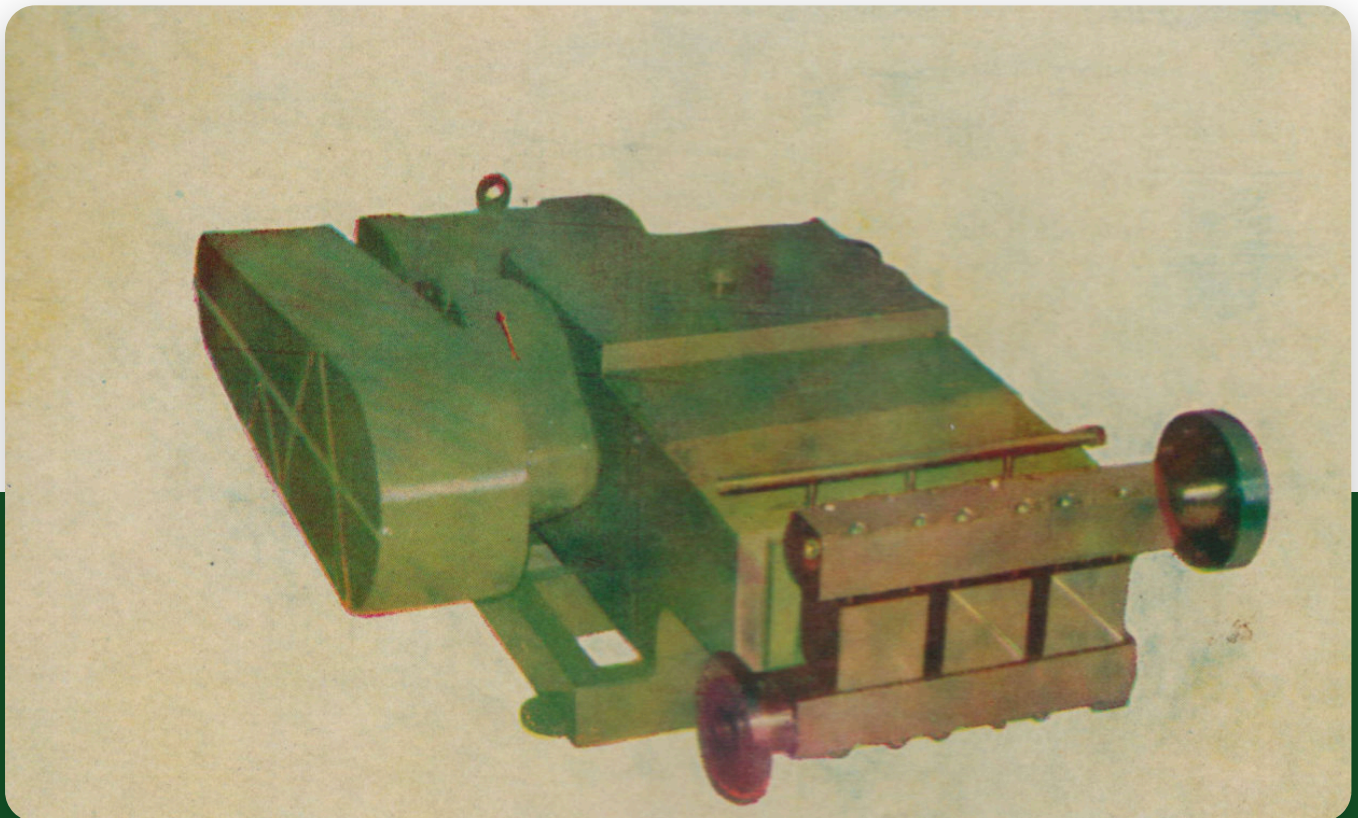
BHAGYODAY
PUMPS
PVT. LTD.



PRODUCT CATALOGUE · SINCE 1991

High-pressure plunger, metering & test pumps

Triplex & quintuplex reciprocating plunger pumps · Metering pumps & dosing systems
· Diaphragm pumps · Motorised hydraulic test pumps



01
Triplex / Quintuplex
Plunger Pumps

02
Triplex Metering Pump

03
Metering Pumps & Dosing
Systems

04
Motorised Hydraulic Test
Pumps

Built for continuous duty

Since 1991, BHAGYODAY engineers and technicians have designed and manufactured high-pressure, low-maintenance reciprocating pumps under the **UDAY** brand, from our works in Naroda, Ahmedabad.

OUR PROMISE

Every pump that bears the UDAY name is crafted from carefully selected materials, painstakingly. Strict quality control at BHAGYODAY is an added guarantee that UDAY pumps give satisfactory service, hour after hour, under real field conditions.

Before shipment, each pump is fully tested and approved to perform to the customer's exact requirements. Liquid ends, materials and accessories can be engineered to suit the liquid being handled.

700

kg/cm² max. discharge pressure (triplex)

1,907

litres / min max. flow (UT-20000)

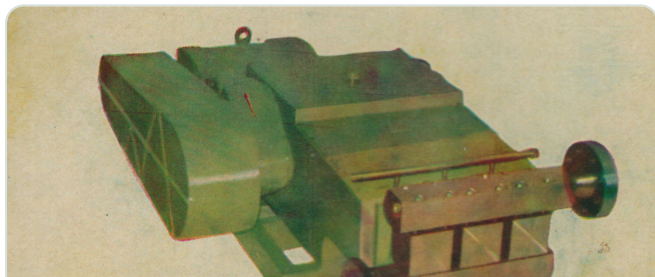
15,000

litres / hr max. metering flow (BPPL 5000)

±1%

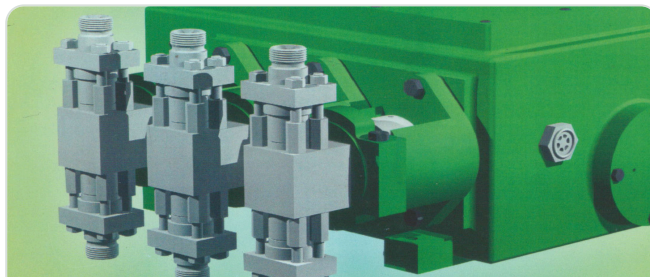
metering accuracy

PRODUCT RANGE



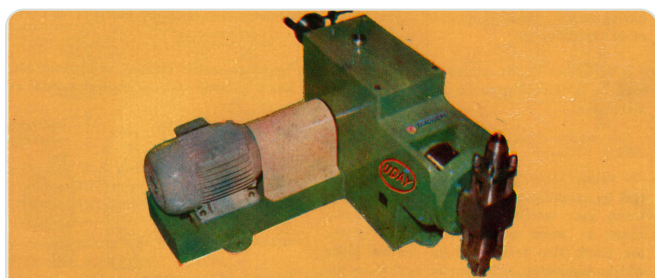
Triplex / Quintuplex Plunger Pumps

UT series · 6 models · up to 700 kg/cm² and 1,907 l/min



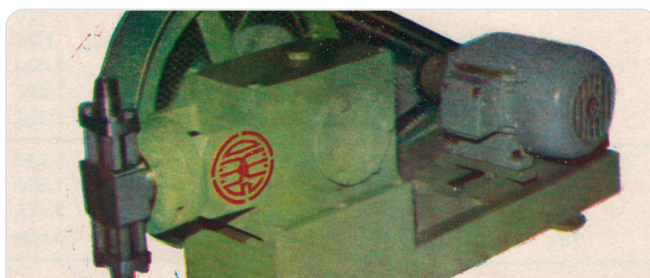
Triplex Metering Pump

Up to three dosing heads on a single prime mover



Metering Pumps & Dosing Systems

BPPL series · plunger & diaphragm heads · 0-100% stroke



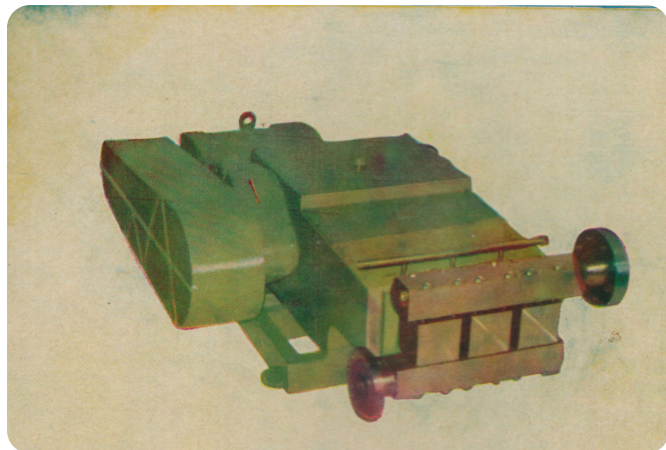
Motorised Hydraulic Test Pumps

Well Test UFF-30 series · portable · up to 120 kg/cm²

We also offer diaphragm pumps, fix-flow pumps and complete packaged chemical dosing systems.

High-pressure reciprocating plunger pumps

Unique high-pressure, low-maintenance, continuous-duty triplex and quintuplex plunger pumps.



APPLICATIONS

- Chemical injection
- Detergent slurry
- Boiler feed
- Process water disposal
- Hydro-jetting & cutting
- Vehicle wash
- Hydrostatic testing
- Reverse osmosis
- Cement grouting
- Salt water injection
- Mechanical separation
- Ammonia scrubber
- ...and many other applications

POWER END

For every UDAY pump the power end is engineered from selected materials and designed for the rated horsepower.

Crank case

Sturdy, completely sealed design in mild steel.

Connecting rods

Accurately machined, self-lubricated, from high-grade material.

Cross head

Large, full piston-type cross head for accurate plunger alignment and minimal side thrust.

Crank shaft

Heavy-duty alloy steel, accurately machined, hardened and ground.

Lubrication

Combined pressure lubrication to crank shaft and cross heads.

Bearings

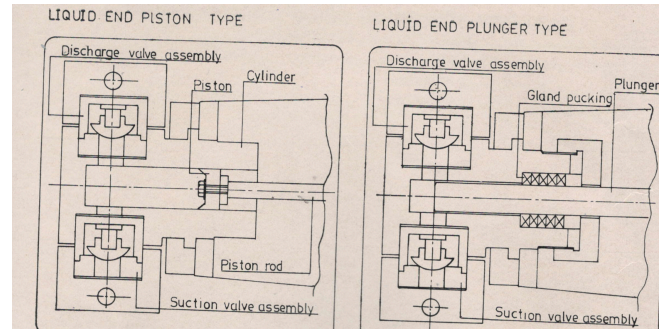
High-quality internal tapered roller bearings support the crank shaft at both ends.

Liquid end & materials

Every UDAY liquid end is painstakingly crafted. BPPL also offers liquid ends tailor-made from a variety of corrosion- and fatigue-resistant materials to match each customer's needs.

Diaphragm-type liquid ends are also supplied for toxic and hazardous liquids.

Valves are available in three types to suit the application: **poppet**, **ball** and **disc**. Surface-treated plungers are fitted as standard for longer life.



Liquid end: piston type (left) and plunger type (right)

ACCESSORIES

- Flanges
- Relief valve
- Back-pressure valve
- Pulsation dampener
- Suction tank
- Jetting gun
- Nozzles
- High-pressure valve
- Trolley

STANDARD MATERIALS OF CONSTRUCTION

Columns are named by liquid-end body material.

Part	Steel	SS 304	SS 316	Aluminium Bronze	SS 316 L
Liquid end body	Steel	SS 304	SS 316	Aluminium Bronze	SS 316 L
Valve	SS 410	SS 304	SS 316	SS	SS 316 L
Valve seats	SS 410	SS 304	SS 316	SS	SS 316 L
Valve spring	SS 304	SS 304	SS 304	SS	SS
Plunger	SS 410	SS 304	SS 316	SS	SS 316 L
Gland	SS 410	SS 304	SS 316	SS	SS 316 L
Gland packing	PTFE / Asbestos rubber	PTFE / Rubber asbestos	PTFE / Rubber asbestos	PTFE / Rubber asbestos	PTFE / Rubber asbestos
'O' ring	Teflon	Teflon	Teflon	Teflon	Teflon

1. Parts in special materials such as Alloy 20, Monel, Hastelloy etc. can be manufactured against specific needs.
2. Ceramic-coated, hard-chrome-plated or nitrided plungers can be supplied.
3. Gland packing flushing, cooling and lubrication can be provided against specific orders.

Capacity selection guide — UT series

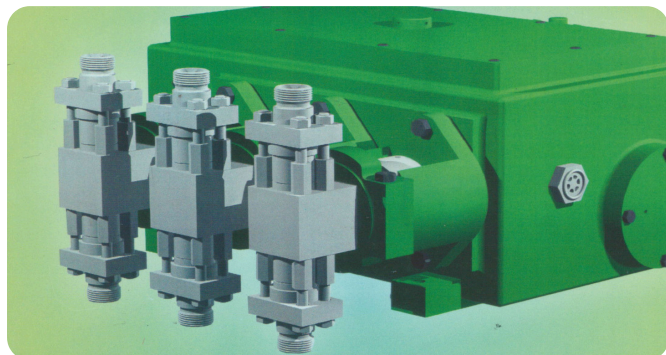
Flow rate in litres per minute at pump speed (RPM). Pressure is the maximum discharge pressure at maximum RPM capacity.

Model	Plunger dia. (mm)	Max. pressure (kg/cm ²)	Flow rate (litres / min)				
			100 RPM	150 RPM	200 RPM	300 RPM	400 RPM
UT-2500	10	700	0.60	0.89	1.27	1.90	2.60
	15	340	1.45	2.15	2.90	4.30	5.80
	20	105	2.50	3.75	5.00	7.50	10.00
	25	125	4.00	6.00	8.00	12.00	16.00
	30	100	5.30	8.00	10.70	16.15	21.53
UT-4000	20	350	3.50	5.00	7.00	10.00	13.00
	25	220	5.00	8.00	10.00	16.00	21.00
	30	150	7.50	11.00	15.00	22.00	30.00
	35	110	10.50	15.50	21.00	31.00	42.00
	40	85	13.50	20.00	27.00	40.00	54.00
UT-5000	25	440	7.00	11.00	14.00	22.00	29.00
	30	300	10.50	15.50	21.00	31.00	42.00
	40	170	18.50	27.50	37.00	55.00	74.00
	50	100	29.00	43.50	58.00	87.00	116.00
	55	90	35.50	53.00	71.00	106.00	142.00
UT-7000	30	500	13.00	20.00	26.00	40.00	53.00
	40	270	23.50	35.50	47.00	71.00	95.00
	50	180	37.00	55.50	74.00	111.00	148.00
	55	150	45.00	67.50	90.00	135.00	180.00
	60	120	53.00	79.00	106.00	158.00	212.00
UT-10000	50	300	53.00	79.00	106.00	158.00	212.00
	60	235	76.00	114.00	152.00	228.00	304.00
	65	198	90.00	135.00	180.00	270.00	360.00
	70	170	103.50	155.00	207.00	310.00	414.00
	75	150	119.00	178.50	238.00	357.00	476.00
	80	132	135.00	202.50	270.00	405.00	540.00
UT-20000	62	200	162.00	243.00	325.00	—	—
	82	180	316.50	474.00	633.00	—	—
	102	160	440.00	660.00	880.00	—	—
	120	140	610.00	915.00	1,220.00	—	—
	140	120	830.00	1,245.00	1,660.00	—	—
	150	100	953.50	1,430.00	1,907.00	—	—

UT-20000 is rated up to 200 RPM. Quintuplex versions and other plunger sizes are available on request.

UDAY triplex metering pump

Ideal for almost any liquid: corrosive chemicals, viscous liquids, oils, slurries, cryogenic liquids, and liquids with high melting points or high temperatures.



UDAY dosing systems are sturdy and precise, suited to continuous duty under harsh conditions with minimum maintenance.

POWER END MECHANISM

The power end converts high-speed rotary motion into slow, accurate reciprocating action. Plunger stroke length is adjustable by hand from 0 to 100%, whether the pump is stopped or running. Flow rate is set by micrometric manual adjustment.

LIQUID HEADS

Wet ends are made from standard or special materials, chosen for the physical and chemical properties of the liquid. Parts are interchangeable across metering duties. Plunger type with single or double ball valve assembly.

VALVES

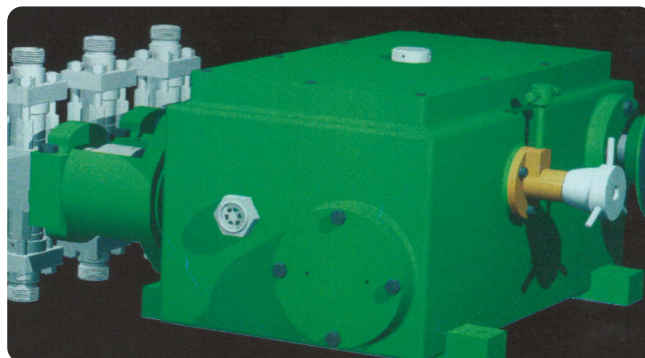
Standardised valve designs for dosing accuracy: single and double ball valve, poppet valve with or without spring loading, and flat valve.

PACKAGED DOSING SYSTEMS

Self-contained units with metering pumps, tanks, stirrers, strainers, instruments, pipes and valves, skid or trolley mounted, or designed to your exact requirement.

SPECIAL FEATURES

- Compact, robust and light-weight
- Accurate over a wide capacity range
- Totally enclosed, water- and dust-proof housing
- Fully automatic lubrication
- Wide range of heads in carefully selected materials
- Multiplexing: up to three heads of different type and capacity on a single prime mover
- Easy availability of spares
- Every pump fully tested before shipment

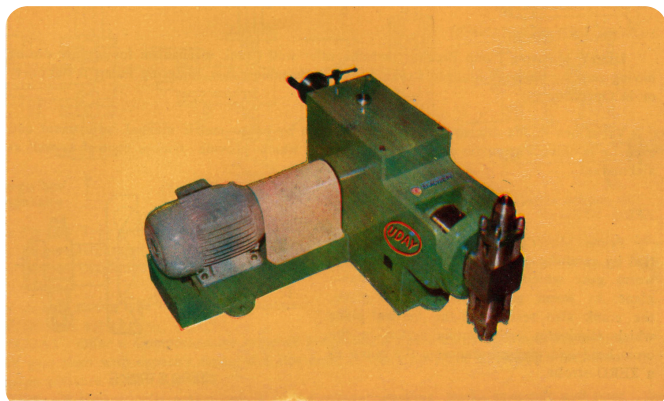


ACCESSORIES

- Relief valve
- Flanged connection
- Pulsation dampener
- Back-pressure regulator
- Suction strainers
- Trolley / skid

UDAY metering pumps & dosing systems

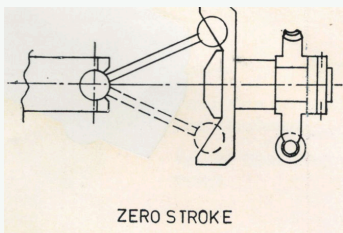
Controlled-volume plunger and diaphragm metering pumps with a polar-crank mechanism for accurate dosing.



POWER END MECHANISM

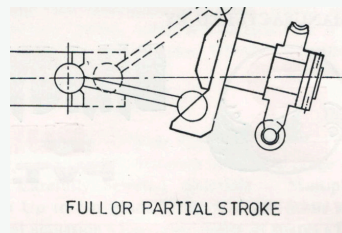
A precision-made high-speed worm drives a worm gear that rotates on an adjustable plane. Plunger stroke length can be set anywhere from **zero to 100%**, with the pump stopped or running. Flow is set by micrometric manual adjustment, or remotely through pneumatic or electric actuators for automation.

Changing the radius of the reciprocating crank gives near-harmonic plunger motion at every stroke length. The suction and discharge ball checks respond in step, so the pump runs efficiently.



ZERO STROKE

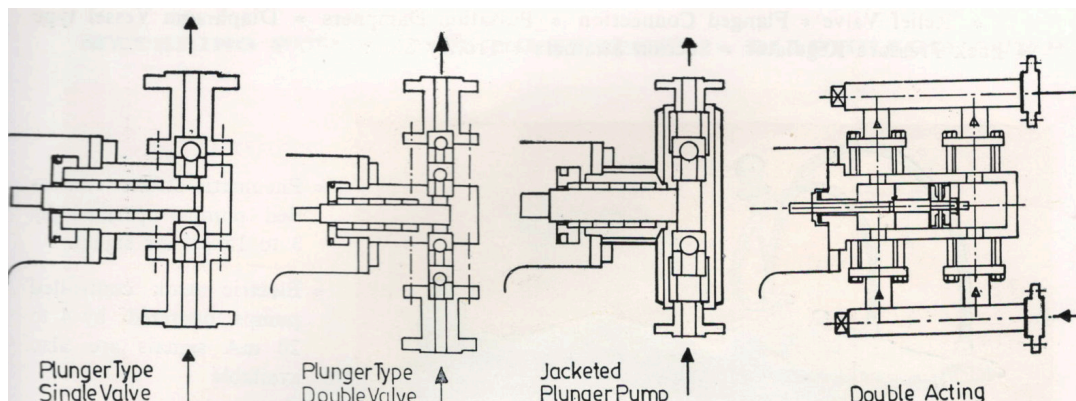
Zero stroke. The worm gear sits in the vertical plane, so the crank turns without moving the cross head or plunger.



FULL OR PARTIAL STROKE

Full or partial stroke. The worm gear is set off vertical, and the connecting rod drives the cross head and plunger.

LIQUID HEADS



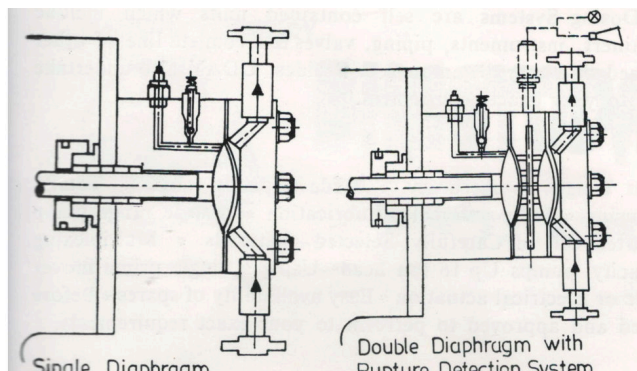
- **Plunger type** with single or double valve assembly
- **Jacketed plunger heads**
- **Single diaphragm** for toxic liquids
- **Double diaphragm** for highly toxic liquids, with optional rupture alarm
- **Double acting** for large capacities

Diaphragm pumps, automation & systems

DIAPHRAGM METERING PUMPS

UDAY offers a full range of hydraulically flexed diaphragm metering pumps for all gear sizes, from a few cc/hr to over **15,000 LPH**, at discharge pressures up to **385 kg/cm²**.

They are hermetically sealed: the plunger packing works in clean hydraulic fluid, and a rugged diaphragm separates it from the metered liquid. Double and sandwiched diaphragm designs with a rupture indicator stop the hydraulic fluid from contaminating the metered liquid.



Single diaphragm (left) and double diaphragm with rupture detection (right)

ADVANTAGES

- Hermetically sealed
- Relief and replenishing valves in the hydraulic circuit
- Better resistance to fouling
- Long life from the PTFE diaphragm and gentle flow against it

AUTOMATION

Pneumatic stroke control on a **3–15 psi** air signal. Electric stroke control on a **4–20 mA** signal.

VARIATIONS

- Jacketed diaphragm heads
- Stainless steel diaphragms
- Remote-head diaphragm pumps

VALVES

Single and double ball valve, poppet valve with or without spring loading, and flat valve.

PACKAGED CHEMICAL DOSING SYSTEMS

Self-contained units with metering pumps, tanks, stirrers, strainers, instruments, piping and valves, plus a complete line of accessories. Supplied skid- or trolley-mounted, or designed and engineered to your exact requirement.

SPECIAL FEATURES

- Compact, robust, light-weight and accurate
- Wide capacity range
- Totally enclosed, water- and dust-proof housing
- Fully automatic lubrication
- Multiplexing: up to **ten heads** on a single prime mover
- Remote control by pneumatic or electric actuation
- Easy availability of spares
- Every pump fully tested before shipment

ACCESSORIES

- Relief valve
- Flanged connection
- Pulsation dampeners (diaphragm vessel type)
- Back-pressure regulator
- Suction strainers
- Trolley / skid

Capacity chart — BPPL 1125 to 2250

Flow rate in litres per hour at 100 and 150 strokes per minute (SPM).

Model	Plunger (mm)	Flow rate (l/hr)		Discharge pressure (kg/cm ²)	Std. connection (BSP)
		100 SPM	150 SPM		
BPPL 1125	3	0.76	1.13	400	
	5	2.00	3.00	400	
	8	5.40	8.00	238	
	12	12.00	18.00	106	½"
	20	34.00	51.00	37	
	30	76.00	114.00	16	
	40	135.00	203.00	9	
	50	210.00	315.00	8	
	55	255.00	380.00	5	¾"
	60	320.00	480.00	4	
BPPL 1750	3	0.95	1.40	700	
	5	2.60	3.90	700	
	8	6.75	5.70	500	
	12	15.00	23.00	136	½"
	20	42.00	63.00	85	
	30	95.00	143.00	38	
	42	186.00	280.00	19	
	50	265.00	397.00	13	¾"
	62	430.00	645.00	7	
BPPL 2250	10	17	25	400	
	12	24	36	276	½"
	20	68	101	101	
	30	152	228	44	
	40	270	406	25	¾"
	50	424	635	16	
	56	530	796	12	
	60	610	914	11	1"
	70	830	1244	8	
	76	980	1466	6	1½"

NOTES

1. Intermediate capacities are available in all models.
2. Alternative stroking speeds of 34, 64, 72 and 150 SPM are also available.
3. Capacities assume 90% volumetric efficiency and may change with discharge pressure.
4. Connection sizes are listed against the plunger size where they appear on the original chart.
5. We reserve the right to change this data without notice.

Capacity chart — BPPL 3000 to 5000

Flow rate in litres per hour at 100 and 150 strokes per minute (SPM).

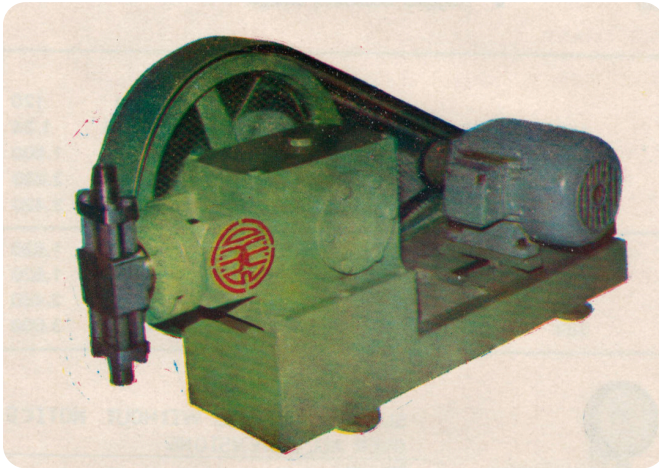
Model	Plunger (mm)	Flow rate (l/hr)		Discharge pressure (kg/cm ²)	Std. connection (BSP)
		100 SPM	150 SPM		
BPPL 3000	12	30	46	580	
	16	54	81	330	½"
	20	85	127	210	
	30	133	286	90	
	40	340	508	53	¾"
	50	530	795	34	
	62	815	1222	22	1"
	75	1192	1788	15	
	80	1356	2034	13	
	90	1716	2575	10	1½"
	100	2119	3180	8	
BPPL 4000	25	165	246	200	
	30	236	355	140	¾"
	40	420	630	75	
	50	657	985	48	1"
	60	946	1419	34	
	70	1288	1931	25	
	80	1682	2523	18	1½"
	90	2128	3193	15	
	100	2628	3942	12	
	120	3784	5676	8	2"
BPPL 5000	25	198	298	350	
	40	508	763	136	1"
	50	795	1192	85	
	70	1558	2336	45	1½"
	85	2297	3445	34	
	100	3179	4768	20	
	125	4967	7451	13	2"
	150	7153	10729	9	
	177	9736	15000	7	2½"

NOTES

1. Intermediate capacities are available in all models.
2. Alternative stroking speeds of 34, 64, 72 and 150 SPM are also available.
3. Capacities assume 90% volumetric efficiency and may change with discharge pressure.
4. Connection sizes are listed against the plunger size where they appear on the original chart.
5. We reserve the right to change this data without notice.

Well Test UFF-30 motorised test pumps

Portable, easy-to-operate test pumps backed by BPPL's experience since 1991: a low-cost pump for accurate hydrostatic pressure.



APPLICATIONS

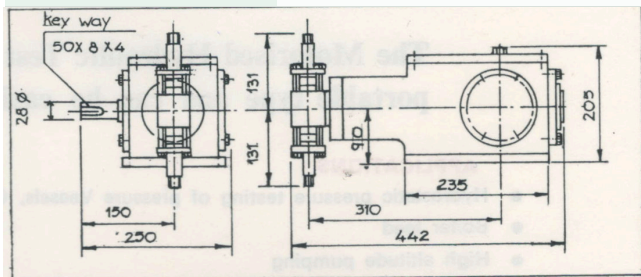
- Hydrostatic pressure testing of pressure vessels, castings, tubes & valves
- Boiler feed
- High-altitude pumping
- Circulating water under pressure

FEATURES

- Totally enclosed
- Easy to maintain
- Oil-bath lubrication for trouble-free operation
- Rugged standard design

PERFORMANCE CHART

Model	LPH	RPM	Power required (BHP) at pressure (kg/cm ²)			
			30	60	90	120
UFF-30/25	150	200	0.310	0.620	0.935	1.240
	225	300	0.400	0.800	1.200	1.600
	300	400	0.508	1.080	1.520	2.030
	375	500	0.605	1.210	1.810	2.420
UFF-30/30	252	200	0.380	0.760	1.140	1.520
	330	300	0.480	0.960	1.440	1.920
	450	400	0.620	1.240	1.860	2.480
	560	500	0.750	1.500	2.250	3.000



Overall dimensions (mm). Subject to change without notice.

SUPPLIED WITH

- Common base plate
- Check valve
- Relief valve
- Pressure gauge

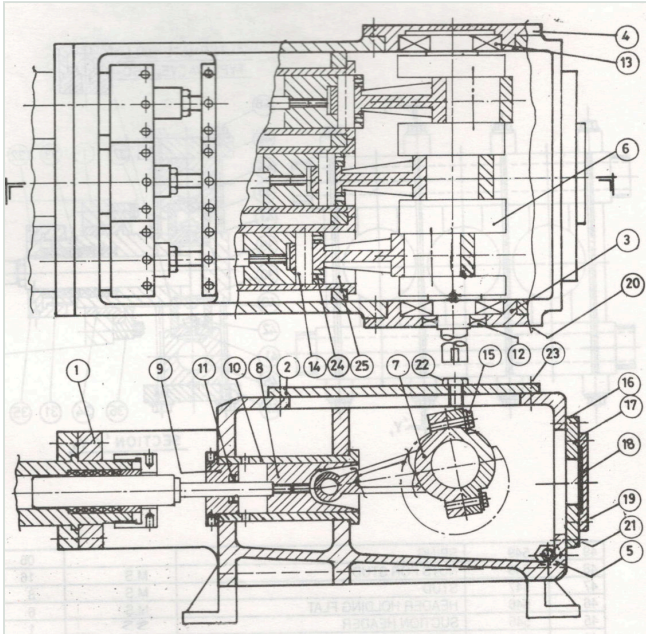
ACCESSORIES

- Suction tank
- Suction / discharge hose
- Skid
- Suction strainer
- Flanges
- Trolley with swivelling caster wheels

Pumps with higher pressure and flow rate are available on request.

Triplex plunger pump — power end assembly

Quote the part code when ordering spares.

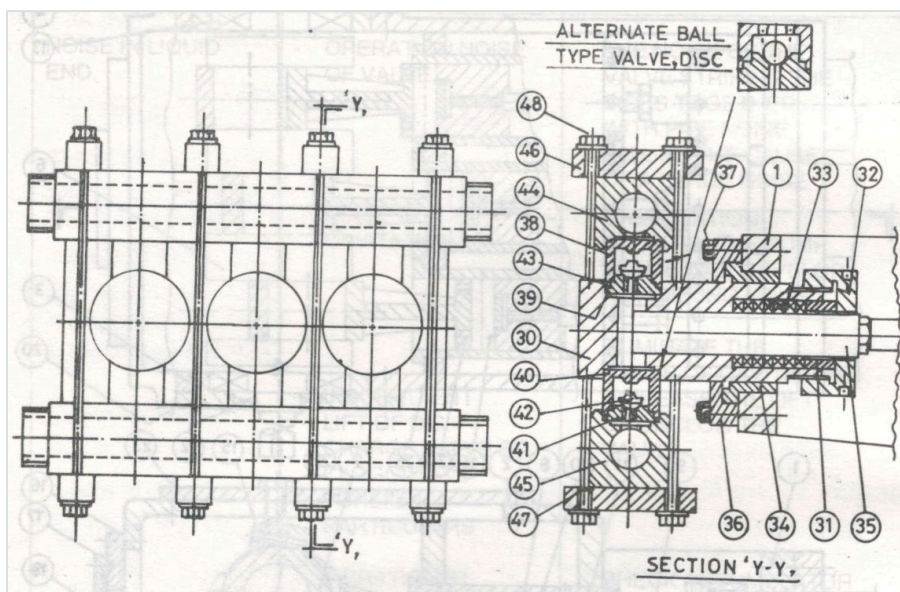


Sectional arrangement, power end

Sr.	Code	Description	Material	Qty
01	501	Main body	C.I.	1
02	502	Top cover	C.I.	1
03	503	Bearing cover (I)	C.I.	1
04	504	Bearing cover (II)	C.I.	1
05	505	Drain plug	M.S.	1
06	506	Crank shaft	EN24	1
07	507	Connecting rod	STD	3
08	508	Piston	C.S.	3
09	509	Sleeve nut	C.I.	3
10	510	Sleeve	C.I.	3
11	511	Plunger side oil seal		3
12	512	Bearing cover side oil seal		1
13	513	Crank shaft bearing		2
14	514	Piston pin	STD	3
15	515	Allen pin for connecting rod	UNbr	6
16	516	Window cover	C.I.	1
17	517	Gauge glass cover	C.I.	1
18	518	Gauge glass	PL.	1
19	519	Allen pin for gauge glass cover	STD	1
20	520	Hex bolt for bearing cover	STD	12
21	521	Hex bolt for window cover	STD	8
22	522	Air plug	C.S.	1
23	523	Hex bolt for top cover	STD	12
24	524	Gazanpin lock	STD	6

Triplex plunger pump — liquid end assembly

Materials shown are standard. Liquid-end parts can be supplied in alternative materials (see Materials of construction).



Sectional arrangement, liquid end (alternate ball-type or disc valve shown inset)

Sr.	Code	Description	Material	Qty
01	501	Main body	C.I.	1
30	530	Liquid end body (block)	SS	3
31	531	Gland	SS	3
32	532	Gland nut	SS	3
33	533	Gland packing	PTFE	3 sets
34	534	Round collar	C.I.	3
35	535	Plunger	SS	3
36	536	Liquid end holding flat	M.S.	3
37	537	Hex bolt for liquid end holding flat	SS	6
38	538	Discharge valve seat (II)	SS	3
39	539	Discharge valve seat (I)	SS	3
40	540	Suction valve seat (II)	SS	3
41	541	Suction valve seat (I)	SS	3
42	542	Valve	SS	6
43	543	Valve seat gasket	Teflon	12
44	544	Discharge header	SS	1
45	545	Suction header	SS	1
46	546	Header holding flat	M.S.	6
47	547	Stud	M.S.	8
48	548	Nuts for stud	M.S.	16
49	549	Spring	—	6

Installation & care essentials

A summary of the UDAY operating and service manuals. Please hand the full manual to your maintenance engineer.

INSTALLATION

- Set the pump level, on a firm common base about 5 cm above the floor, as close to the supply as possible.
- Ensure **positive suction**. Triplex pumps need at least 2 m of water head at ambient temperature.
- Use suction pipe of full inlet size, or the next size up for long runs. It must be completely air-tight, with long-radius bends.
- Size piping for peak flow, about 3.14× the average for reciprocating pumps.
- Fit a correctly sized **safety valve** between the pump and the discharge shut-off valve, venting back to the storage tank.
- Fit shut-off valves on both the suction and discharge sides.

STARTING A NEW PUMP

- Pumps are dispatched **without oil**. Open and clean the crank case, drain any water, then fill to the indicator line with the correct grade. Do not overfill.
- Check that the direction of rotation matches the arrow on the pump.
- Run slowly at near-zero discharge pressure for several hours, then build up gradually to full speed and pressure.
- Watch for heating, abnormal noise, vibration or suction air leaks.

ROUTINE MAINTENANCE

- Change the oil after the first **100 hours**, then every **500 hours**.
- Tighten the gland gradually, a quarter turn every 10 minutes on triplex pumps or 1/8 turn on metering pumps. Overtightening damages the packing and plunger.
- Inspect valves and seats regularly for wear, coatings or particles.
- Metering pumps: calibrate capacity at 100%, 50% and 10% stroke under actual operating conditions.

RECOMMENDED LUBRICANTS — TRIPLEX PUMPS

Maker	Heavy duty	Medium duty	Light duty
Hindustan Petroleum	Enklo 460	Enklo 320	Enklo 220
Bharat Petroleum	Hydrol 460	Hydrol 320	Hydrol 220
Indian Oil	Servo 460	Servo 320	Servo 220

Metering pumps use worm-gear oil. Medium grade suits normal duty. Avoid lubricants with strong EP additives. See the metering pump manual for grades.

WARRANTY

BPPL warrants products of its own manufacture against defects in material and workmanship for **18 months from supply or 12 months from commissioning**, whichever is earlier, when installed, operated and maintained as instructed. Installation and transport costs are not included. Performance guarantees are subject to a variation of 3%.



BHAGYODAY
PUMPS
PVT. LTD.



Since 1991, manufacturers of **UDAY** high-pressure triplex & quintuplex plunger pumps, metering pumps, dosing systems, diaphragm pumps, fix-flow pumps and hydraulic test pumps.

WORKS & OFFICE

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– 382 330, Gujarat, India

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bhagyodaypumpspvtltd@gmail.com

CIN

U29120GJ1991PTC015802

INCORPORATED

10 June 1991 · ROC Ahmedabad

SENDING AN ENQUIRY?

Please include: liquid handled · flow rate · discharge pressure · suction conditions · operating temperature · material preference · drive / power supply.

Specifications are subject to change without notice.